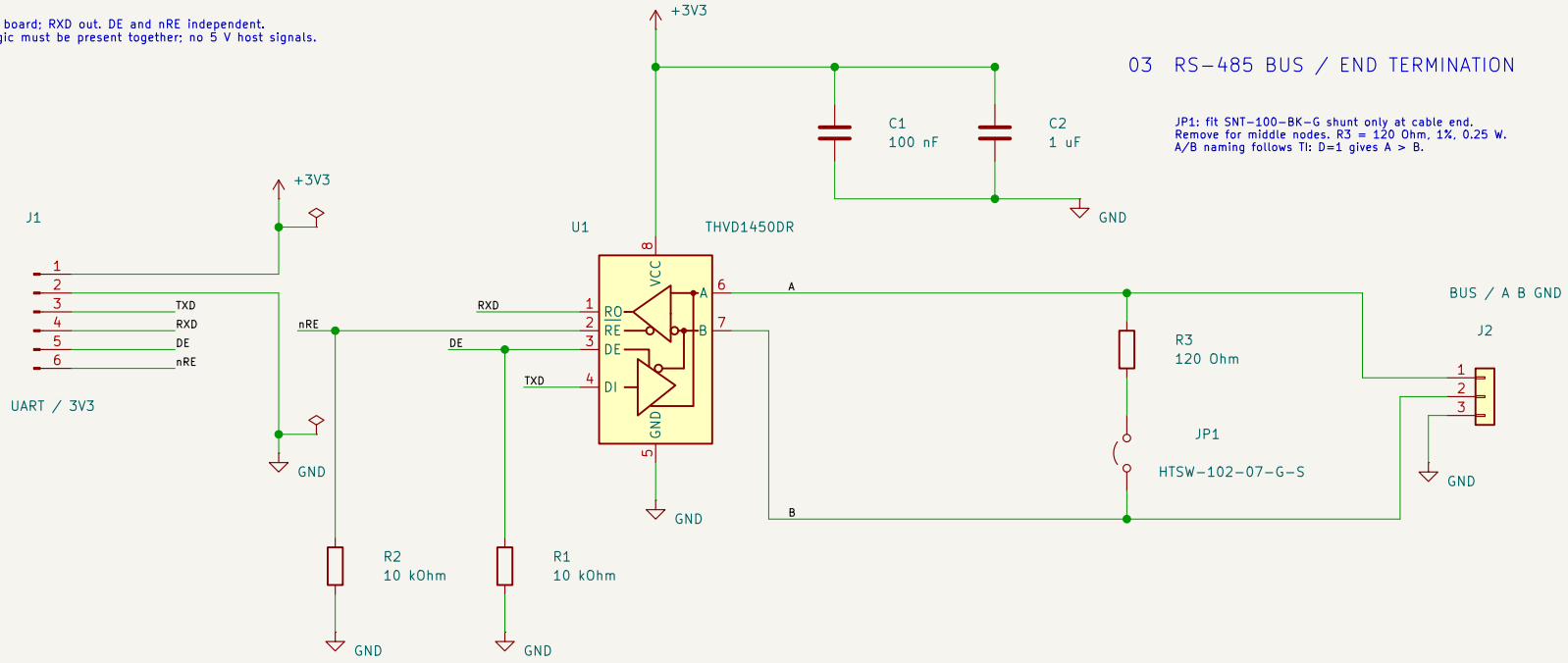
01 HOST UART / 3.3 V ONLY

J1: TXD into board; RXD out. DE and nRE independent.
Supply + logic must be present together; no 5 V host signals.

02 TRANSCEIVER + LOCAL BYPASS

C1 at U1 pin 8. C2 adds board-entry reservoir; not a TI minimum.
THVD1450DR: 1=RO 2=nRE 3=DE 4=DI 5=GND 6=A 7=B 8=VCC



03 RS-485 BUS / END TERMINATION

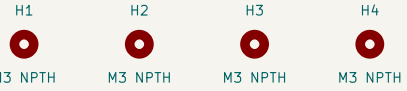
JP1: fit SNT-100-BK-G shunt only at cable end.
Remove for middle nodes. R3 = 120 Ohm, 1%, 0.25 W.
A/B naming follows TI: D=1 gives A > B.

DEFAULT: DE=0 disables drive; nRE=0 enables receive.
10k max 10.1k x 6.2uA = 62.62mV < VIL 0.8V (TI).
Controller leakage and startup state still need review.

DESIGN BASIS / UNBUILT DRAFT

3.3V nominal / 3.0-3.6V input. Initial bench target 115200 baud.
No isolation, external TVS, surge, EMC or link-speed qualification.
TI internal failsafe: no external A/B bias fitted.
R3 bound: 3.6V^2 / 118.8 Ohm = 109.1mW; verify real bus faults.

MECHANICAL / FABRICATED



4 x 3.2mm NPTH / 50 x 35mm PCB
No purchased fasteners selected.

SOURCES + REVISION

A1 / 2026-10-29: initial educational interface. TI SLLSEY3E: fig31,40; sec9.4.11,12.
Exact manufacturer bytes + pin/footprint decisions: evidence/index.yaml, DESIGN_BASIS.md.
Author visual inspection is not owner signoff. Native ERC/DRC is not hardware validation.

THVD1450DR; termination only at physical cable ends
UNBUILT / NON-ISOLATED / NO HARDWARE QUALIFICATION
PCB Copilot educational examples

Sheet: /		
File: rs485-interface.kicad_sch		
Title: RS-485 interface | 3.3 V UART / half-duplex		
Size: A4	Date: 2026-10-29	**Rev: A1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/1